# Hierarchical Household-Level Matching & Generative Models

## Preserving Intra-Household Coordination for Activity Schedule Generation

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# Outline

Context & Literature

Current Framework: EQASIM Individual Matching

Our Approach: Hierarchical Household Matching

Experimental Setup

Results

Generative Models: Household CVAE Evolution

# Context & Literature

# What is an Activity Chain ? What is it for ?

## 1. The Input: Daily Activity Chains

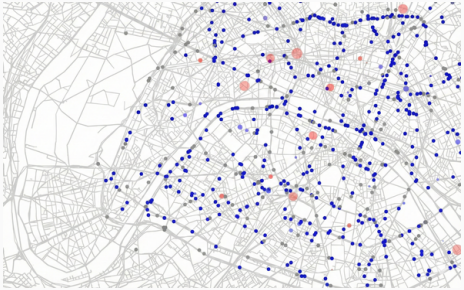



↓ *Numerous agents execute their schedules simultaneously* ↓

## 2. The Output: Dynamic Simulation

**MATSim**

**Transport simulation pipelines** need realistic daily activity schedules for synthetic populations **to generate accurate microscopic travel demand.**

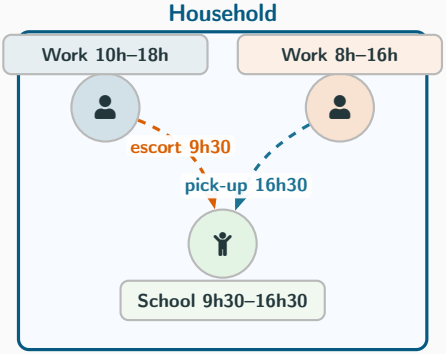


# Why Household-Level?

Problem: Individual-level matching/generation **ignores** coordination between household members:

- Shared car usage
- School drop-offs (escort trips)
- Joint shopping / leisure

**Goal:** Match/Generate entire synthetic households trips based off HTS data while preserving these joint behaviors.



# Literature Overview & Baseline

**TASHA (Roorda)**

Rule-based heuristic + Monte Carlo sampling. Priority scheduling: core act. → joint act. → flexible indiv act.

⚠ Sequential bias + Underestimation of flexible act.

**OASIS (Pougala/Hillel/Revzany)**

Parallel MiLP on collective random utility. Hard constraints to implement joint participation + vehicle allocation. MH for choice set generation.

⚠ Hard to implement + Underestimation of flexible act.

**RAG-LLM (Liao 2024)**

GPT-4o with RAG intra-household. Prompt-based coordination. Limited info setting.

⚠ Special case + average results

**ActVAE (Shone/Hillel 2025)**

Continuous CVAE + LSTM. Individual level only. Best at capturing behavioral variability.

✓ Reproducible and good results
✗ Only individual level

**DeepCAM (Liao 2025)**

Transformer encoder-decoder with cross-attention & role embeddings. Sequential: leader → others.

✓ Good aggregate stats
⚠ Heavy implementation

**EQASIM (Hörl 2021)**

Individual-level statistical matching. Weighted CDF sampling + progressive relaxation.

✓ Reproducible, operational-focused
✗ Destroys joint & escort trips

**Ours: Hierarchical Matching + Generative CVAE**

1. Household-level matching as an upper bound to strictly preserve observed joint coordination.
2. Iterations of a Household CVAE to enable schedule generation and explicitly capture interactions.

✓ Preserves exact structural joint behaviors / generates coherent individual variations.

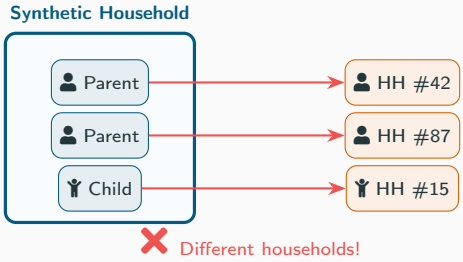
First part of the internship

# Current Framework: EQASIM Individual Matching

# EQASIM: Individual-Level Statistical Matching

**How it works:**

1. Match each individual independently
2. Attributes: sex, age class, any_cars, CSP, département
3. Weighted CDF sampling (person weights from HTS)
4. Progressive constraint relaxation ($5 \rightarrow 0$ attributes)




**Pros:**

- Fast and efficient matching
- Ensures diversity through random offsets

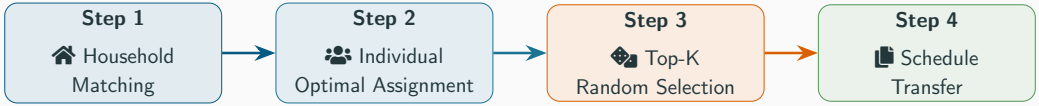
**Cons:**

- Members of same household $\neq$ same donor household
- Joint trips / joint activities / escort trips destroyed

First part of the internship

# Our Approach: Hierarchical Household Matching

# Algorithm Overview



# Step 1: Household-Level Candidate Selection

**Exact matching** on household attributes:

- household_size
- number_of_vehicles
- departement_id
- income_class
- urban_type

Must be **strictly identical** between target & donor

**Progressive relaxation** if $|\Omega_h| < K_{\min}$:

Level 0: All 5 variables

↓

Level 1: Drop `departement_id`

↓

Level 2: Drop `urban_type`

↓

Level 3: size + vehicles only

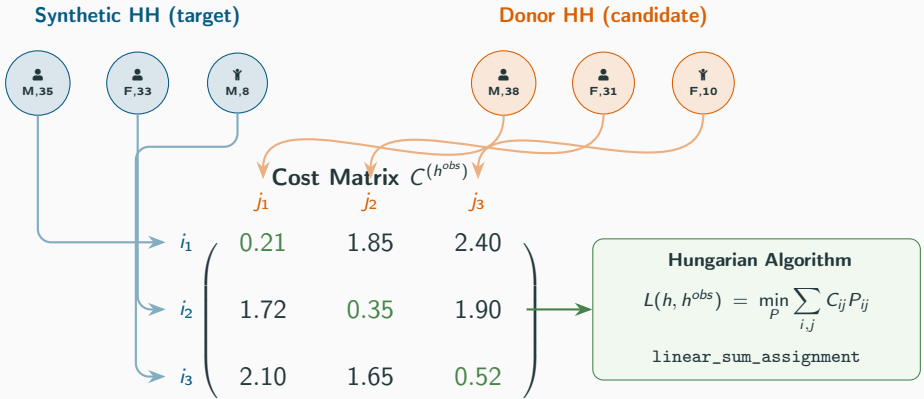
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Level 4: size only

if not enough candidates

Candidates **accumulate** across levels until $\geq K_{\min}$ reached.

# Step 2: Individual-Level Cost Matrix

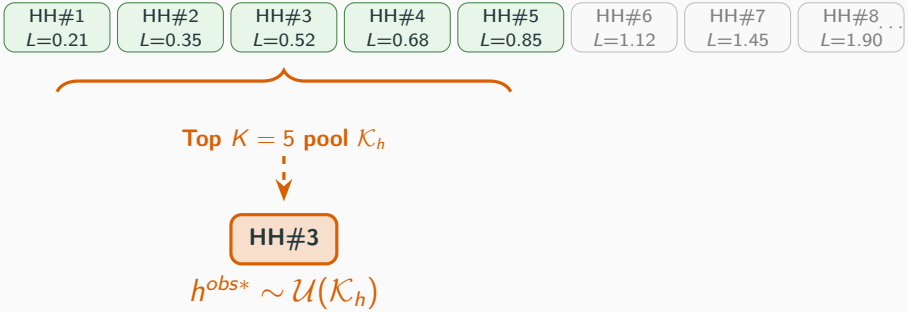



**Distance between individuals $u$ and $v$:**

$$\text{dist}(u, v) = \lambda_{\text{age}} \cdot \underbrace{\frac{|c_u - c_v|}{K}}_{\text{ordinal}} + \lambda_{\text{cat}} \cdot \underbrace{\sum_k \mathbb{1}(u_k \neq v_k)}_{\text{nominal}}$$

# Step 3: Stochastic Top-K Selection

**All candidates ranked by $L(h, h^{obs})$**




**Why not just take the best?** Uniform sampling among top-$K$ preserves natural behavioral variance across simulation runs. Deterministic → every synthetic household gets the same donor → reduced diversity.

# Experimental Setup

# Data & Setup

**Source (HTS):**

- EGT 2010 (Enquête Globale Transport)
  - 14 736 households, 34 580 individuals
  - Full trip chains + spatial zones (200m grid)

**Target (Synthetic Population):**

- EQASIM pipeline (0.31% sampling)
  - 14 414 households, 32 419 individuals
  - Census-based attributes

**Matching Variables:**

**Household Level**

- household_size
- number_of_vehicles
- income_class (10 bins)
- urban_type
- departement_id

**Individual Level**

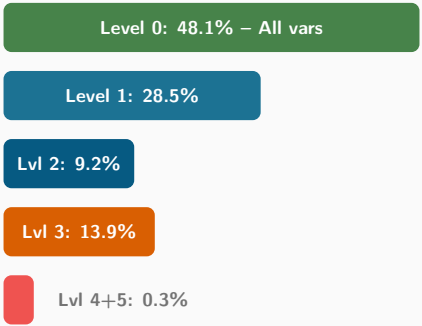
- age (9 classes)
- sex, employed, studies
- CSP, commute_mode

$K_{\min} = 15$, *Top-K* $= 5$, *10 runs for statistics.*

# Results

# Matching Quality

## Relaxation Distribution:




**76.6%** matched with $\geq$4 variables (levels 0–1)

## Candidate Pool Statistics:

| Statistic | Value |
|---|---|
| Mean candidates / HH | 129.8 |
| Median | 42.0 |
| Min – Max | 2 – 2000+ |
| Unmatched | 1 household |

Atypical profiles (especially large HH) have few candidates $\rightarrow$ relaxation needed.

# Aggregate Evaluation: BASELINE vs EQASIM

| Theme | Metric | EQASIM | BASELINE (Ours) |
|---|---|---|---|
| Participation | Avg Activity Count EMD | 2.71% | 4.76% |
| Transition | 2-gram $L_1$ | 6.4% | 13.3% |
| | 3-gram $L_1$ | 8.1% | 16.4% |
| Timing | Start Time EMD (min) | 4.39 | 5.17 |
| | Duration EMD (min) | 5.11 | 7.21 |
| Diversity | Real Seq. Coverage | 61.5% | 51.4% |

⚠ **Expected trade-off**: Forcing all members to come from the same donor household **excludes potentially great individual matches** from the wider pool → slightly worse individual-level metrics.

# Household-Level Evaluation: Joint Behaviors 🤝

**Joint Activity Proportions:**

| | EQASIM | BASELINE |
|---|---|---|
| Joint Activities | 0.18% | ✅ 10.6% |
| Joint Trips | 0.46% | ✅ 11.3% |
| Escort Trips | 0.89% | ✅ 3.9% |
| *EGT Reference* | *10.7% / 11.2% / 4.3%* | |

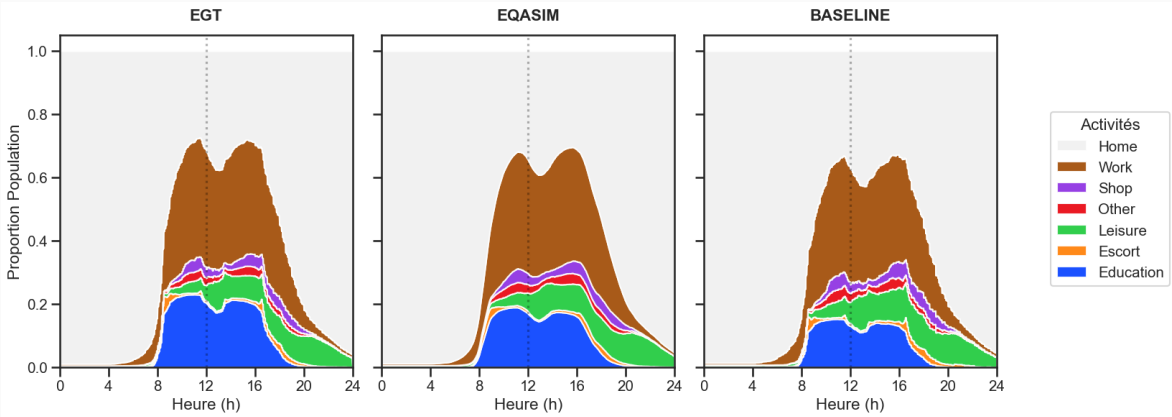
**EMD on Joint Behaviors (min):**

| Metric | EQASIM | BASELINE |
|---|---|---|
| JA Duration | 17.57 | ✅ $16.45 \pm 1.12$ |
| JA Start Time | 84.90 | ✅ $9.94 \pm 1.25$ |
| JT Start Time | 153.85 | ✅ $15.73 \pm 1.39$ |
| Escort Start | 105.62 | ✅ $12.34 \pm 2.10$ |

🎯 BASELINE reproduces ∼**98%** of the EGT's joint activity rate!

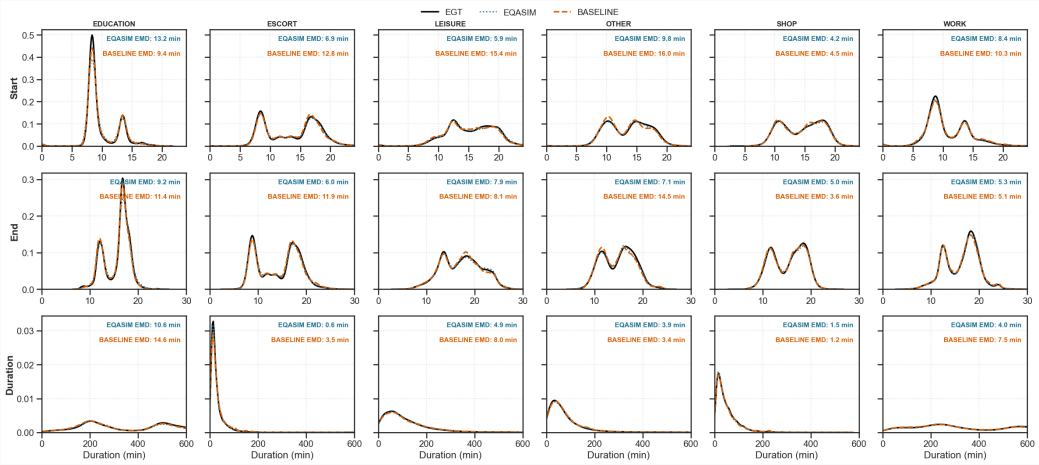
⚠️ EQASIM metrics are skewed by extremely low sample counts. BASELINE delivers viable timings on a realistic volume.

# Visual: Activity Participation Over 24h



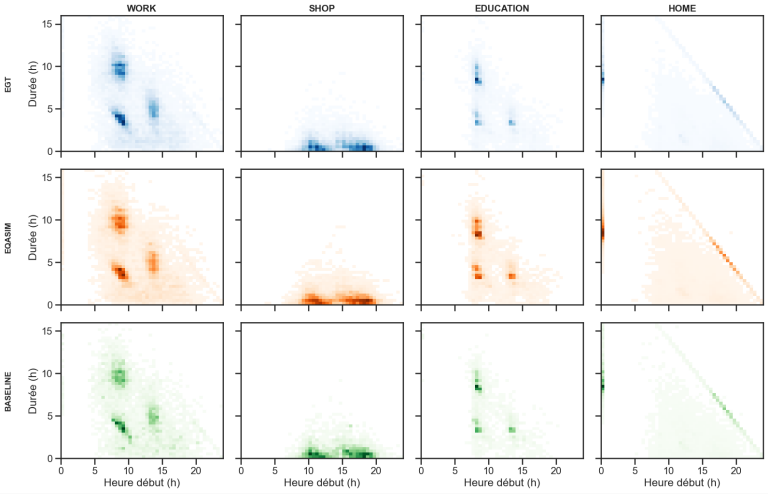

- The 24h activity profile of BASELINE closely matches the EGT reference distribution.
- Underestimation of education due to matching young adults with students/children.

# Marginal Distributions by Activity Purpose




The generated temporal distributions (Start, End, Duration) remain close to the EGT reference across all activity types for both models.

# Joint Distributions: Start Time vs. Duration




Our approach accurately reproduces the **joint distributions** of start times and durations,

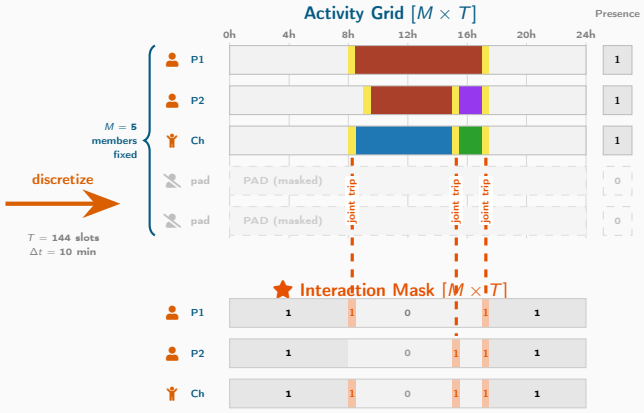
Second part of the internship

# Generative Models: Household CVAE Evolution

# Data Encoding: From Activity Episodes to Tensors

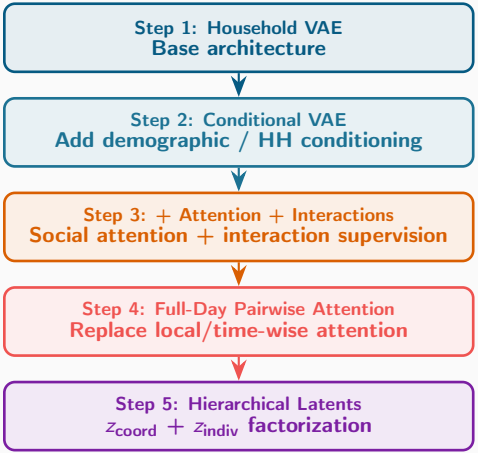
## Trip Data (EGT)

| pid | act | start | end |
|---|---|---|---|
| P1 | Home | 0 | 480 |
| P1 | Trip | 480 | 510 |
| P1 | Work | 510 | 1020 |
| P1 | Trip | 1020 | 1050 |
| P1 | Home | 1050 | 1440 |
| P2 | Home | 0 | 540 |
| P2 | Trip | 540 | 570 |
| P2 | Work | 570 | 900 |
| P2 | Trip | 900 | 930 |
| P2 | Shop | 930 | 1020 |
| P2 | Trip | 1020 | 1050 |
| P2 | Home | 1050 | 1440 |
| Ch | Home | 0 | 480 |
| Ch | Trip | 480 | 510 |
| Ch | Education | 510 | 900 |
| Ch | Trip | 900 | 930 |
| Ch | Leisure | 930 | 1020 |
| Ch | Trip | 1020 | 1050 |
| Ch | Home | 1050 | 1440 |




**Interaction mask** derived from overlapping schedules:
if two or more members share the same activity at the same time and at the same place → interaction flag = 1.

# Household CVAE – Architecture Evolution Roadmap

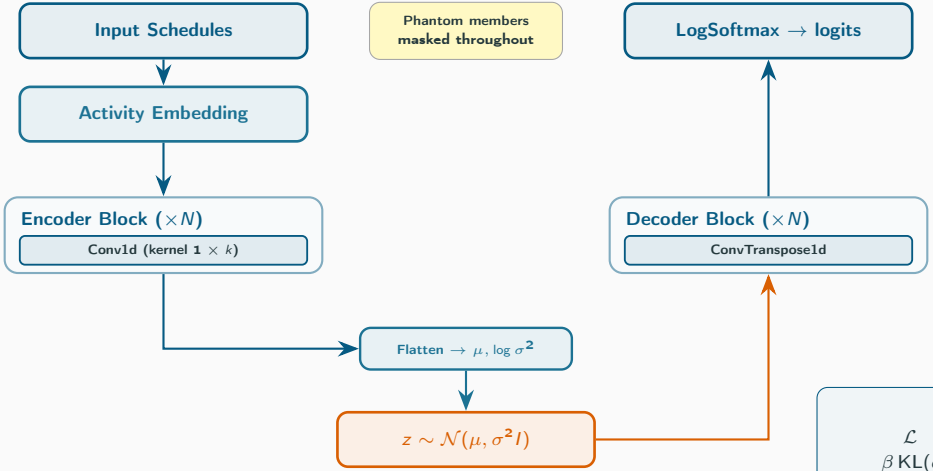



## Design logic

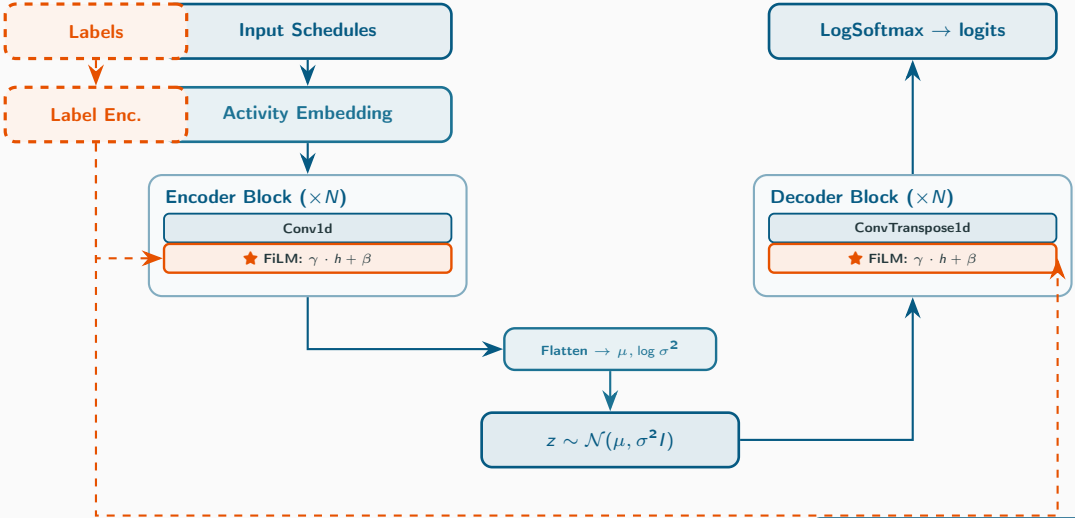
- Start simple: Try to reconstruct daily schedules
- Add modulable **conditioning** for attributes
- Add **explicit** interaction supervision and **implicit** attention to capture social behaviors
- Try to improve **social coupling** via full-day attention → global visibility of all members' schedules
- Factorize latent space into shared vs individual variability

**Dashed orange** = new or modified vs previous step.

# Step 1: Household VAE (Base)



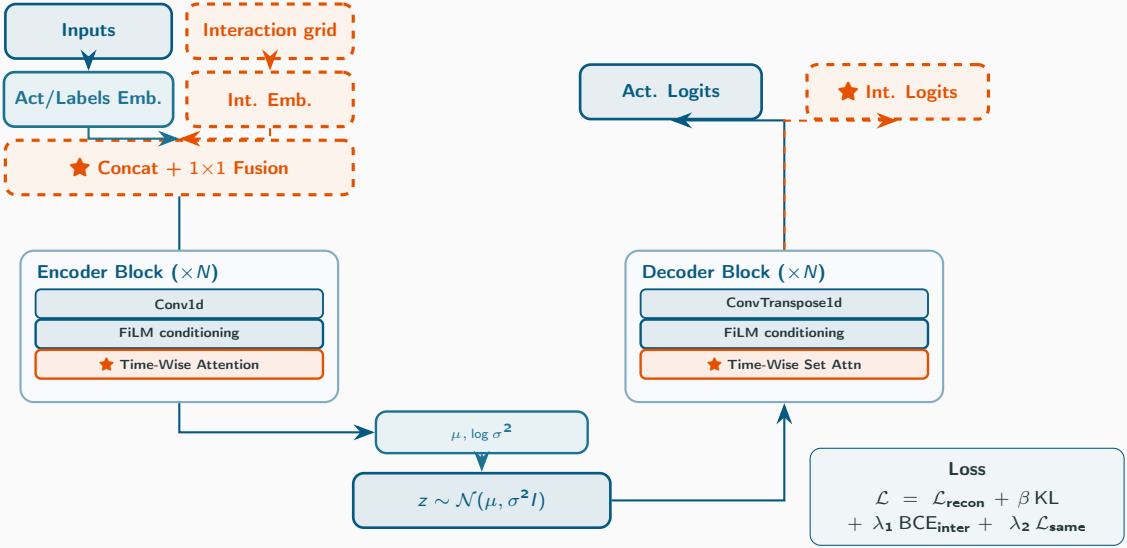
# Step 2: Conditional VAE with FiLM



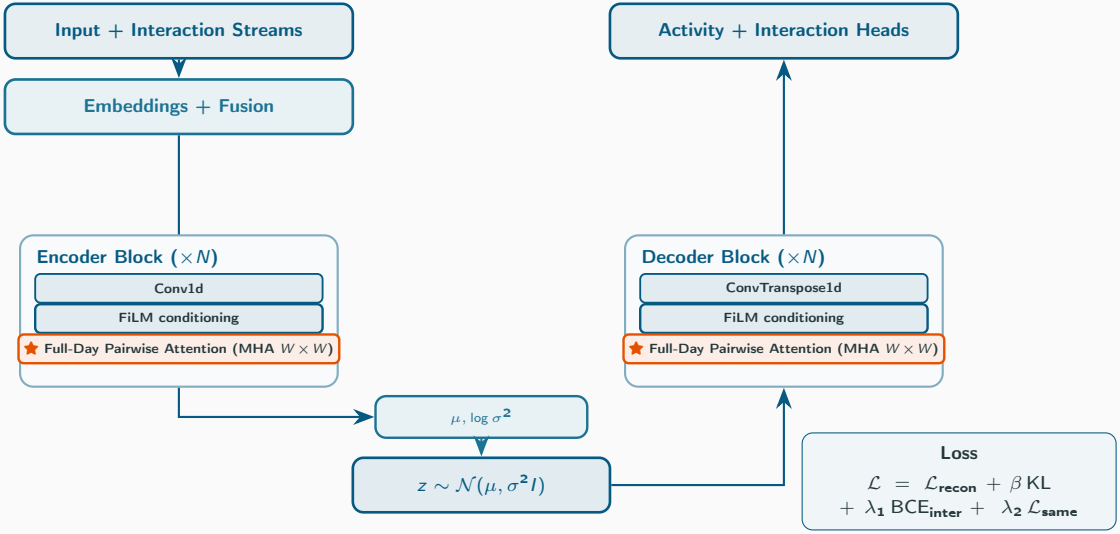

**Loss**

$$\mathcal{L} = \mathcal{L}_{\text{recon}} + \beta\,\text{KL}$$

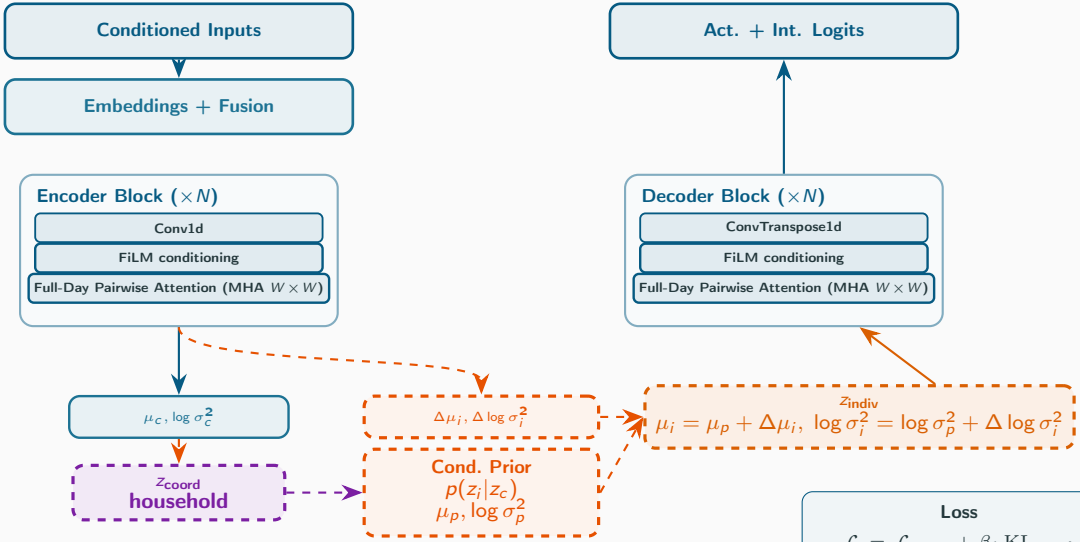
# Step 3: + Interactions + Time-Wise Social Attention



# Step 4: Full-Day Pairwise Attention



# Step 5: Hierarchical Latents ($z_{\text{coord}} + z_{\text{indiv}}$)




**Loss**

$$\mathcal{L} = \mathcal{L}_{\text{recon}} + \beta_1 \,\mathrm{KL}_{\text{coord}} + \beta_2 \,\mathrm{KL}_{\text{indiv}} + \lambda\, \mathcal{L}_{\text{social}}$$

# Generative Models: Aggregate Metrics Comparison

| Theme | Metric | DiscCNN | VAE | CVAE | Att | FullAtt | FullHier |
|---|---|---|---|---|---|---|---|
| Participation | Avg Act Count EMD | 0.279 | 0.309 | 0.292 | **0.267** | 0.306 | 0.341 |
| Timing | Duration EMD (min) | 40.23 | 53.61 | 51.79 | 26.68 | **25.02** | 44.97 |
| | Start Time EMD (min) | **26.48** | 39.79 | 37.81 | 27.54 | 32.14 | 47.83 |
| Transition | 2-gram $L_1$ | 0.325 | 0.304 | **0.286** | 0.386 | 0.432 | 0.514 |
| | 3-gram $L_1$ | 0.634 | 0.641 | **0.574** | 0.636 | 0.766 | 0.835 |
| | **Interaction Behavior EMD (min)** – *Interaction-capable models only* | | | | | | |
| Interaction | JA Duration | – | – | – | **22.14** | 44.74 | 93.80 |
| | JA Start Time | – | – | – | **20.04** | 40.45 | 75.24 |
| | JT Start Time | – | – | – | **61.76** | 80.16 | 230.51 |

And a near-neutral novelty profile (closest train/test bias: 0.00–0.03), suggesting very limited training-set memorization compared to the baseline's 0.88 score.

# Generative Models: Example of Generated Schedules

